> *RGA is most useful as a stress-response mechanism: it preserves the static attention path when reliability evidence is high, and it helps under coherent score-collapse conditions when reliability evidence indicates domain failure.*

The measured evidence supports this claim on the secondary benchmark. The clean table is near saturated and should be treated as a sanity check. The reliability-stress experiments, the threshold sweep, and the component ablation provide the more useful evidence about the mechanism.

1.3 Contributions

We make six scoped contributions:

1. We present ELARA, a score-level anomaly-fusion research system with a unified long-format schema for heterogeneous evidence.
2. We define RGA, a conservative reliability-gated attention module that combines masked attention with calibration, drift, and sharpness signals.
3. We evaluate the method against practical score-fusion baselines, including random forest fusion, early-fusion MLP, late-fusion ensemble, confidence-weighted mean, score-level Tent, and pseudo-label TTT.
4. We report an all-category paired MVTec 3D-AD benchmark, multi-seed confidence intervals, mechanism ablations, failure cases, and explicit threats to validity.
5. We add the D13 natural positive-transfer track, which requires a validation-only candidate to beat both SAR and confidence-weighted mean on a fresh natural holdout; opened 3D-ADAM and MulSen results are retained as development evidence only.
6. We add the D16 Real-IAD D3 natural-degradation headroom audit, which evaluates a validation-only stress selector on natural degraded samples and reports the official stress-subset fail separately from the supportive all-test CW result.

1.4 Evidence Boundary

The evidence boundary is part of the contribution because it prevents the results from being overstated. The real-domain benchmark uses real local datasets, but the records are label-aligned. The MVTec benchmark uses natural RGB/3D pairing across eight categories, but it still uses lightweight normal-reference image statistics and the canonical one-class anomaly protocol. These two artifacts answer different questions. The first is useful for reliability-stress score fusion. The second is useful for exposing protocol limits on naturally paired data. Neither should be treated as a complete final benchmark for all multimodal anomaly detection.

1.5 Dataset Use Matrix

Table 1 records how the local datasets are used in this chapter. The table is included because the repository contains official evidence, development evidence, diagnostic proxy datasets, and utility scaffolding in the same tree. Only prelocked naturally paired data can carry a transfer claim; opened